

EJERCICIOS ARPWATCH

EJERCICIOS SENCILLOS OBLIGATORIOS

Ejercicio 1 — Instalar arpwat

Objetivo: tener la herramienta lista.

sudo apt update

sudo apt install -y arpwat

```
(kali@kali)~$ sudo apt update
[sudo] password for kali:
Get:1 https://repo.protonvpn.com/debian stable InRelease [2,967 B]
Get:2 https://repo.protonvpn.com/debian stable/main amd64 Packages [8,313 B]
Get:3 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:4 https://repo.protonvpn.com/debian stable/main all Packages [207 kB]
Get:5 http://kali.download/kali kali-rolling/main amd64 Packages [20.7 MB]
Get:6 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.1 MB]
Get:7 http://kali.download/kali kali-rolling/contrib amd64 Packages [117 kB]
Get:8 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [268 kB]
Get:9 http://kali.download/kali kali-rolling/non-free amd64 Packages [190 kB]
Get:10 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [905 kB]
Get:11 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [11.8 kB]
Get:12 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [30.0 kB]
Fetched 74.5 MB in 6s (12.9 MB/s)
2120 packages can be upgraded. Run 'apt list --upgradable' to see them.
arpwatch is already the newest version (2:1a15-9).
The following packages were automatically installed and are no longer required:
  libbluray2 libgdata-common libgeos3.13.1 libjs-underscore libplacebo349 libqt5ct-common1.8 libsigsegv2 libsoup2.4-common libtheoradec1 libudfread0 python3-packaging-whl
  libgda136 libgdata22 libhdf4-0-alt libogdi4.1 libportmidi0 libsfml1 libsoup-2.4-1 libtheora0 libtheoraenc1 libvpx9 python3-wheel-whl
Use 'sudo apt autoremove' to remove them.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2120
```

✓ **Qué comprobar:** que la instalación termina sin errores.

♦ Ejercicio 2 — Ver la interfaz de red

Objetivo: identificar la interfaz que se va a vigilar.

ip addr show

✓ **Qué anotar:** el nombre de la interfaz (ej.: eth0, enp0s3, wlan0).

```
(kali@kali)-[~]
$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d1:f8:5d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 81341sec preferred_lft 81341sec
    inet6 fd17:625c:f037:2:8e8a:a9f1:b99a:7bb6/64 scope global dynamic noprefixroute
        valid_lft 86288sec preferred_lft 14288sec
    inet6 fe80::e91:d06b:115:7c7a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:31:97:65 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.126/23 brd 192.168.1.255 scope global dynamic noprefixroute eth1
        valid_lft 38442sec preferred_lft 38442sec
    inet6 fe80::aea9:6c35:9b9a:8014/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

♦ Ejercicio 3 — Arrancar arpwat

Objetivo: poner en marcha el programa.

`sudo arpwat -i eth0 -d`

```
(kali@kali)-[~]
$ sudo arpwat -i eth1 -d
```

Si no escucha:

`sudo touch /var/lib/arpwat/arp.dat`

`sudo chown arpwat:arpwat /var/lib/arpwat/arp.dat`

(reemplaza eth0 por la interfaz que hayas visto en el Ejercicio 2).

✓ **Qué verás:** la consola queda “escuchando” la red.

♦ Ejercicio 4 — Generar un evento

Objetivo: que arpwat detecte algo nuevo.

- Abre **otra terminal** y haz ping a cualquier dirección de tu red:
- `ping -c 3 192.168.0.1`

```
(kali㉿kali)-[~]  
$ sudo arpwatch -i eth1 -d  
  
From: arpwatch (Arpwatch kali)  
To: root  
Subject: new station (192.168.0.41) eth1  
  
      hostname: <unknown>  
      ip address: 192.168.0.41  
      interface: eth1  
ethernet address: 02:7c:dc:38:c8:10  
ethernet vendor: <unknown>  
      timestamp: Monday, January 26, 2026 3:55:19 -0500
```

✓ **Qué verás:** en la terminal de arpwatch aparecerá una línea con new station (IP + MAC).

♦ Ejercicio 5 — Ver los logs

Objetivo: comprobar que queda registrado el evento.

`sudo journalctl -f | grep arpwatch`

```
(kali㉿kali)-[~]
└─$ sudo journalctl -f | grep arpwatch
Jan 26 04:09:10 kali sudo[9219]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/arpwatch
ch -i eth1 -d
Jan 26 04:09:10 kali arpwatch[9222]: listening on eth1
Jan 26 04:09:33 kali arpwatch[9222]: exiting
```

Si no saliese:

`sudo mkdir -p /var/lib/arpwatch`

`sudo touch /var/lib/arpwatch/arp.dat`

`sudo chown -R arpwatch:arpwatch /var/lib/arpwatch`

```
(root㉿kali)-[/home/kali]
# cd /var/lib/arpwatch

(root㉿kali)-[/var/lib/arpwatch]
# ls
arp.dat  arp.dat-  ethercodes.db

(root㉿kali)-[/var/lib/arpwatch]
# cat arp.dat
08:2e:5f:09:05:38      192.168.0.81      1769417714      eth1
08:ed:ed:b6:33:cc      192.168.0.72      1769417782      eth1
a4:97:33:4e:9f:c4      192.168.0.90      1769417801      eth1
08:00:27:31:97:65      192.168.0.126     1769417788      kali eth1
08:2e:5f:2f:f1:74      192.168.0.43      1769417656      eth1
6c:3b:e5:40:2c:2e      192.168.0.77      1769417600      eth1
08:00:27:0c:00:75      192.168.0.44      1769417608      eth1
00:0c:29:13:85:b3      192.168.0.209     1769417613      eth1
ce:e5:38:5d:75:26      192.168.0.110     1769417613      eth1
18:60:24:e9:3b:2a      192.168.0.23      1769417648      eth1
78:ac:c0:bd:a1:59      192.168.0.125     1769417664      eth1
08:ed:ed:b5:aa:e6      192.168.0.68      1769417803      eth1
08:2e:5f:09:05:51      192.168.0.46      1769417670      eth1
02:7c:dc:38:c8:10      192.168.0.41      1769417794      eth1
2c:44:fd:27:f1:32      192.168.0.123     1769417756      eth1
6c:3b:e5:0e:91:2d      192.168.0.85      1769417771      eth1
```

EJERCICIOS AVANZADOS OPCIONALES

Ejercicio 1 — Instalar arpwat

Objetivo: tener la herramienta disponible.

`sudo apt update`

`sudo apt install -y arpwat`

`which arpwat`

`arpwat -v`

```
(kali@kali)-[~]
$ which arpwat
/usr/sbin/arpwat

(kali@kali)-[~]
$ arpwat -v
Version 2.1a15
usage: arpwat [-dN] [-f datafile] [-F "filter"] [-i interface] [-n net[/width]] [-r file] [-s sendmail_path] [-p] [-a] [-m addr] [-u username] [-Q] [-z ignorenet/ignoremask]
```

Observa: ruta del binario y versión. Si no aparece, revisa errores de instalación.

Ejercicio 2 — Identificar la interfaz de red

Objetivo: saber qué interfaz monitorizar.

`ip link show`

o

`ip addr show`

```
(kali@kali)-[~]
$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d1:f8:5d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 79661sec preferred_lft 79661sec
    inet6 fd17:625c:f037:2:8e8a:a9f1:b99a:7bb6/64 scope global dynamic noprefixroute
        valid_lft 86023sec preferred_lft 14023sec
    inet6 fe80::e91:d06b:115:7c7a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:31:97:65 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.126/23 brd 192.168.1.255 scope global dynamic noprefixroute eth1
        valid_lft 36761sec preferred_lft 36761sec
    inet6 fe80::aea9:6c35:9b9a:8014/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Observa: nombre de la interfaz (p. ej. eth0, enp0s3, wlan0). Usa ese nombre en los siguientes ejercicios.

Ejercicio 3 — Arrancar arpwatc

Objetivo: ver la salida directa para entender su funcionamiento.

`sudo arpwatc -i eth0 -d`

```
(kali㉿kali)-[~]
└─$ sudo arpwatc -i eth1 -d

From: arpwatc (Arpwatc kali)
To: root
Subject: new station (192.168.0.28) eth1

        hostname: <unknown>
        ip address: 192.168.0.28
        interface: eth1
        ethernet address: 08:00:27:d3:15:dc
        ethernet vendor: PCS Systemtechnik GmbH
        timestamp: Monday, January 26, 2026 4:16:31 -0500
^C
```

(Reemplaza eth0 por tu interfaz.)

Observa: mensajes en pantalla sobre nuevas estaciones o cambios ARP.

Ejercicio 4 — Ejecutar arpwatc como servicio y comprobar proceso

Objetivo: poner como servicio y confirmarlo.

`# systemd`

`sudo systemctl enable --now arpwatc`

`ps aux | grep arpwatc`

```
(kali㉿kali)-[~]
└─$ sudo systemctl enable --now arpwatc
Synchronizing state of arpwatc.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable arpwatc

(kali㉿kali)-[~]
└─$ ps aux | grep arpwatc
root      8833  0.0  0.3 12344 6452 ?        S   04:01   0:00 arpwatc -i eth1 -u root
root      8854  0.0  0.3 11600 6228 ?        S   04:01   0:00 arpwatc -i eth1
root      8882  0.0  0.3 12344 6744 ?        S   04:02   0:00 arpwatc -i eth1 -u root
kali      9815  0.0  0.1  6528 2276 pts/0    S+  04:26   0:00 grep --color=auto arpwatc
```

`# alternativa si no hay servicio:`

`sudo pkill -f arpwatc; sudo arpwatc -i eth0 -d &`

Observa: la línea en ps con el proceso arpwatc y su PID.

Ejercicio 5 — Monitorizar logs en tiempo real

Objetivo: ver las alertas que genera arpwatrch.

si tu sistema usa syslog

sudo tail -f /var/log/syslog | grep -i arpwatrch

si usa systemd/journal (Kali, etc.)

sudo journalctl -f | grep -i arpwatrch

```
(kali㉿kali)-[~]  
$ sudo journalctl -f | grep -i arpwatrch  
Jan 26 04:27:11 kali arpwatrch[8882]: new station 192.168.0.103 08:00:27:82:39:62 eth1  
Jan 26 04:27:11 kali arpwatrch[9818]: execl: /usr/lib/sendmail: No such file or directory  
Jan 26 04:27:11 kali arpwatrch[8882]: reaper: pid 9818, exit status 1  
█
```

Observa: entradas tipo new station o changed ethernet address.

~~Ejercicio 6 — Conectar un nuevo host (generar evento)~~

~~Objetivo: provocar un evento de “nueva estación” y verlo en logs.~~

- ~~• Levanta una VM extra o conecta un móvil a la misma red/segmento.~~
 - ~~• Observa los logs (usa el comando del ejercicio 5).
Observa: línea new station <IP> <MAC> en los logs.~~
-

Ejercicio 7 — Cambiar la MAC local y comprobar alerta

Objetivo: cambiar temporalmente la MAC y ver la detección.

bajar interfaz

sudo ip link set dev eth0 down

cambiar MAC (ejemplo)

sudo ip link set dev eth0 address 02:11:22:33:44:55

subir interfaz

sudo ip link set dev eth0 up

observar logs

sudo journalctl -f | grep -i arpwatrch

```

(kali㉿kali)-[~]
$ sudo ip link set dev eth1 down

(kali㉿kali)-[~]
$ sudo ip link set dev eth1 address 02:11:22:33:44:55

(kali㉿kali)-[~]
$ sudo ip link set dev eth1 up

(kali㉿kali)-[~]
$ sudo journalctl -f | grep -i arpwatch
Jan 26 04:29:44 kali arpwatch[8882]: new station 192.168.0.50 a0:02:a5:69:81:dd eth1
Jan 26 04:29:44 kali arpwatch[9928]: execl: /usr/lib/sendmail: No such file or directory
Jan 26 04:29:44 kali arpwatch[8882]: reaper: pid 9928, exit status 1
^C

```

(Reemplaza eth0 y la MAC según tu caso.)

Observa: arpwatch registra changed ethernet address para la IP afectada.

Ejercicio 8 — Consultar la base de datos de arpwatch

Objetivo: ver el historial que guarda arpwatch.

```
sudo ls -l /var/lib/arpwatch
```

```
sudo sed -n '1,80p' /var/lib/arpwatch/arp.dat
```

Observa: entradas con IP, MAC y marcas de tiempo.

```

(kali㉿kali)-[~]
$ sudo ls -l /var/lib/arpwatch

[sudo] password for kali:
total 1156
-rw-r--r-- 1 root      root          3055 Jan 26 05:05 arp.dat
-rw-r--r-- 1 root      root          3055 Jan 26 04:35 arp.dat-
-rw-r--r-- 1 arpwatch arpwatch 1172057 Sep 26 03:10 ethercodes.db

(kali㉿kali)-[~]
$ sudo sed -n '1,80p' /var/lib/arpwatch/arp.dat
08:2e:5f:09:05:38      192.168.0.81      1769419868          eth1
08:ed:ed:b6:33:cc      192.168.0.72      1769419854          eth1
a4:97:33:4e:9f:c4      192.168.0.90      1769419843          eth1
08:00:27:31:97:65      192.168.0.126     1769419723          kali   eth1
08:2e:5f:2f:f1:74      192.168.0.43      1769419532          eth1
6c:3b:e5:40:2c:2e      192.168.0.77      1769419636          eth1
08:00:27:0c:00:75      192.168.0.44      1769417608          eth1

```

Ejercicio 9 — Añadir una entrada ARP estática (mitigación)

Objetivo: fijar una IP↔MAC para protegerla en el equipo local.

sintaxis moderna recomendada

```
sudo ip neigh add 192.168.1.143 lladdr 9f:4b:0c:9f:14:cb nud permanent dev eth0
```

comprobar

```
ip neigh show
```

```
(kali㉿kali)-[~]
$ sudo ip neigh add 192.168.1.143 lladdr 9f:4b:0c:9f:14:cb nud permanent dev eth1

(kali㉿kali)-[~]
$ ip neigh show

10.0.2.2 dev eth0 lladdr 52:55:0a:00:02:02 STALE
192.168.1.143 dev eth0 lladdr 9f:4b:0c:9f:14:cb PERMANENT
192.168.1.143 dev eth1 lladdr 9f:4b:0c:9f:14:cb PERMANENT
fd17:625c:f037:2::2 dev eth0 lladdr 52:56:00:00:00:02 router STALE
fe80::2 dev eth0 lladdr 52:56:00:00:00:02 router STALE
```

Observa: que la entrada aparece como PERMANENT o REACHABLE según sistema.

Ejercicio 10 — Parar y limpiar la práctica

Objetivo: dejar el entorno como estaba.

parar arpwatch

```
sudo systemctl stop arpwatch
```

eliminar entrada ARP creada (ejemplo)

```
sudo ip neigh del 192.168.1.143 dev eth0
```

comprobar que ya no existe

```
ip neigh show
```

```
(kali㉿kali)-[~]
$ sudo systemctl stop arpwatch

(kali㉿kali)-[~]
$ sudo ip neigh del 192.168.1.143 dev eth1

(kali㉿kali)-[~]
$ ip neigh show

10.0.2.2 dev eth0 lladdr 52:55:0a:00:02:02 STALE
192.168.1.143 dev eth0 lladdr 9f:4b:0c:9f:14:cb PERMANENT
fd17:625c:f037:2::2 dev eth0 lladdr 52:56:00:00:00:02 router STALE
fe80::2 dev eth0 lladdr 52:56:00:00:00:02 router STALE
```